

## LESSON 6-8 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

# Find the Greatest Common Factor

Builds on Lesson 6-7 Session 1 • 6.NS.B.4

**TODAY'S LESSON**You will factor  $20t + 45$  by pulling out the GCF of 20 and 45.**YOU NEED TO**Find the GCF of two numbers, like 20 and 45: the GCF is 5. **You will use it in Practice Form A, problems 2 and 3.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** List all the **factors** of the first number.
- 2** List all the **factors** of the second number.
- 3** Find the **greatest** number on both lists.

**WATCH ME***the steps in action***Find the GCF of 24 and 30.****Step 1** 24: 1, 2, 3, 4, 6, 8, 12, 24**Step 2** 30: 1, 2, 3, 5, 6, 10, 15, 30**Step 3** Both lists: 1, 2, 3, 6. The greatest is 6.**Answer 6****TRY IT WITH HELP***fill in each step*

- 1.**
- Find the GCF of 12 and 16.

**Step 1** 12: 1, 2, 3, \_\_\_\_\_, 6, 12**Step 2** 16: 1, 2, \_\_\_\_\_, 8, 16**Step 3** Both lists: 1, 2, \_\_\_\_\_. The GCF is \_\_\_\_\_.

- 2.**
- Find the GCF of 14 and 35.

**Step 1** 14: 1, 2, \_\_\_\_\_, 14**Step 2** 35: 1, 5, \_\_\_\_\_, 35**Step 3** Both lists: 1, \_\_\_\_\_. The GCF is \_\_\_\_\_.**YOUR TURN***use the steps on your own*

- 3.**
- Find the GCF of the two numbers in each expression.

**a.**  $8x + 12$  \_\_\_\_\_

**b.**  $10y + 15$  \_\_\_\_\_

**c.**  $6n + 9$  \_\_\_\_\_

**d.**  $14a + 21$  \_\_\_\_\_

**e.**  $20m + 30$  \_\_\_\_\_

**f.**  $9k + 4$  \_\_\_\_\_

# Distribute and Combine Like Terms

Builds on Lesson 6-8 Session 1 • 6.EE.A.3

**TODAY'S LESSON**You will simplify expressions like  $3(x + 10) + 2x$ .**YOU NEED TO**Distribute first, then combine like terms, like  $4(n + 3) + 2n = 6n + 12$ . **You will use it in Practice Form A, problems 4 and 5.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Distribute:** multiply the outside number by each term inside.
- 2 Combine** the variable terms.
- 3 Write** the variable term, then the number.

**WATCH ME***the steps in action***Simplify  $4(n + 3) + 2n$ .****Step 1**  $4n + 12 + 2n$ **Step 2**  $4n + 2n = 6n$ **Step 3**  $6n + 12$ **Answer**  $6n + 12$ **TRY IT WITH HELP***fill in each step***4.** Simplify  $2(x + 3) + 5x$ .**Step 1**  $2x + \underline{\hspace{2cm}} + 5x$ **Step 2**  $2x + 5x = \underline{\hspace{2cm}}$ **Step 3** Write:  $\underline{\hspace{2cm}}$ **5.** Simplify  $6(k + 5) - 2k$ .**Step 1**  $6k + \underline{\hspace{2cm}} - 2k$ **Step 2**  $6k - 2k = \underline{\hspace{2cm}}$ **Step 3** Write:  $\underline{\hspace{2cm}}$ **YOUR TURN***use the steps on your own***6.** Simplify. Distribute first.**a.**  $3(a + 4) + a = \underline{\hspace{2cm}}$ **b.**  $5(m + 1) + 2m = \underline{\hspace{2cm}}$ **c.**  $7(b + 2) + b = \underline{\hspace{2cm}}$ **d.**  $8(r + 1) - 3r = \underline{\hspace{2cm}}$ **e.**  $e + 4(e + 2) = \underline{\hspace{2cm}}$ **f.**  $2(d + 6) + 4d = \underline{\hspace{2cm}}$ **READY CHECK***a problem like the one in today's lesson***7.** Simplify  $2(z + 6) + 3z$ .**A**  $5z + 12$ **B**  $5z + 6$ **C**  $2z + 15$ **D**  $17z$ **CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about #